

ActInfLab 2021 Quarterly Roundtable #3

Roundtable | 1:10:47

hello

everyone welcome back to the active

inference lab

welcome to our second semester today

it's august 3rd

2021 so it's two

cubed i guess nice date stephen please

don't double or mute um

welcome everyone we're really happy that

you're joining us today

in the active inference lab we're a

participatory online lab that is

communicating

learning and practicing applied active

inference

you can find us at the links that are

here on this page

this is recorded in an archived live

stream so please provide us with

feedback so that we can be improving on

our work

all backgrounds and perspectives are

welcome here and we'll be following

video etiquette for live streams with

muting if there's noise in the

background and

raising our hand when we're ready to

speak

today in our third quarterly round table

we're going to

go over the goals and provide sort of an

overview on the lab itself

then we're going to go into updates for

each of the three

organizational units of the lab which

are the dot edu
dot coms and dot tools and uh
give some updates as well as talk about
how people can get involved with each of
those
sections of the lab so we'll start with
our
goals maybe while i do some little tweak
on the stream
alex do you want to go ahead on this
website slide
uh no please go ahead oh sure well this
is our website
activeinference.org and if you go there
now you'll see it
looking much like this as well as
providing a lot more information
as far as links to follow us on and
forms you can
fill out but the goal of active
inference lab is to produce cutting edge
research and enable
real-world applications of active
inference so that's our
inference side and our action side we're
learning by doing and we're doing it
about
active inference so if you want to read
more information about that
check out the website to provide
kind of a architectural overlay for the
active inference lab
this figure shows oh let me
uh yeah this figure shows how the
different organizational units that are
internal to our lab
are related to the interfaces and to
external communities so

i'll give an overview and then anyone
can raise their hand and
give a thought on it the three
organizational units
within the lab that we're going to be
talking more about today are
dot edu dot tools and dot coms
and that's our internal states then
at the interfaces that's where we have
interactions with
external entities so for example this
live stream is a production of dot-coms
it's right on the interface between the
internal and the external states of our
lab
and it interfaces out with broader
communities
so those are people who are learning
active inference or curious about
applying active inference or just what
it is
as well as all kinds of communities that
may or may not realize that they're
adjacent to active inference
dot coms also reaches out to other
groups and
explores how active inference might play
a role in their work
and in edu the production on the
interface
is really the maintenance of the body of
knowledge which again we'll be
talking more about soon but we
take a sort of systems engineering
approach to diagramming our own
activities and also to working
internally and externally
any thoughts on that

yeah scott yeah i think the the value
of structuring the uh research
enterprise and the interaction
enterprise
in a way that reflects the content is
absolutely superb that's something that
um living the
and breathing the active inference i
think is really
going to be instrumental as the i guess
i'll call it the movement grows there's
a lot of desire for deep risking out
there and there's a lot of appeal
in different areas banking food system
security a lot of areas were
starting to introduce the concept and
having us
organize the lab in that way um really
walking the walk
i think is is really terrific and speaks
to the
um flexibility and strength of the
framing in a bunch of two different
organizational constructs so i think
that's something that's
from the start was very wise for the org
for you guys to do as an organization
nice thanks it's almost like if you
understand this org
chart or this layout then you're you're
half the way
there you understand there's internal
and external states and that they
interface
together so any other comments or we can
continue on
to the updates yes stephen
well i suppose the only challenge you

the the perennial challenge
is what's internal what's external
where's the boundary
um and i think that that's
proving to be an evolving conversation
um i think that i think that's never
answered in a way
in a in a complete sense but um
you know is it is it going to be more
like there's multiple ways of thinking
about that if you're doing statistics
if you're doing some teleological
morphogenesis
type approach if you're doing some sort
of
top-down kind of using self-reported
data
and you're making a whole load of
assumptions so i think that's quite
quite quite useful and one thing that
came up and i'm not sure which paper it
was but
i saw um carl first and i think he
mentioned the idea of
something called an approximation
science
which i hadn't heard before but there's
almost a science of making
approximations seem to come into play
here
um i haven't delved into that but that
seems to also
maybe tie into that question
cool indeed another yeah yeah go ahead
scott
i just had another comment on stephen's
comment you know that idea the boundary
is really interesting because

now because the lab uses the structure
we'll be able to identify those parts of
the lab function that are
generative and the ways in which they're
generative
and so that notion stephen about the
boundary and where that resides gets
really interesting then
because the entire framing generates
awareness and so the boundary
in a way it almost is less important not
not that the question's not
the question super interesting but the
actual location of a boundary
is almost less important than the
processes that occur
around a given arbitrary boundary within
because system is going to have multiple
boundaries
and so it's a fascinating question of
the where the round a boundary resides
kind of within a complex of boundaries
all of which are
engaged in active inference
simultaneously in the system
because it's one of the things i'm
understanding
is there's no mono dimensional system
that's beyond a theoretical system your
their systems are running all sorts of
active inference calculations
simultaneously of different weightings
etc
and so the boundary starts to become
like drawing a circle with tangents in a
way drawing a boundary
with tangential active inference
uh dynamics in a way and and you have

multiple

active interfaces going to it so that

was a fascinating the question you just

asked got me thinking about that

um that could boundary construction in a

different way and very dynamic

boundaries that would occur

thanks nice and uh yep many ways of

going about active inference as we'll be

discovering

and we arbitrage that against the

clarity of what active inference lab is

itself like each of these units are

their own

informational blanket there's a drive

folder there's a weekly meeting

so we're sure about that those are not

the ambiguous boundaries but that's

actually how we go about investigating

the real world and the systems that do

have uncertainty that's fundamental

let's go on to the updates so the first

update which happened right at the end

of last semester was that actin flab

has a new steady state as a non-profit

organization

registered in california united states

of america

and this was part of our labs strategy

and medium-term plan right from the

beginning

any thoughts on this alex or we'll have

more to say on this in just a few

slides

um about this

action and these new states important

thing

important thing to think about is the

new affordances
and again we can consider it in a notion
of active inference
what possible opportunities we can get
and you use for
strategy realization some new
contacts and i think we have some
uh information in slides about on some
next slide about
what possible opportunities are
yes one of the
cornerstones and motivators of
our project and lab as well as the
registration as a non-profit especially
is this concept of research debt
and alex how about you maybe summarize
what
research debt means in this context
[Music]
in general the notion of research depth
uh
is about some needed
uh and special actions what need to be
done
for some new state-of-the-art research
uh because people who are doing uh
really
great research in their fields spending
very time to get to wet frontiers but
after that to force
such fresh knowledge to be accessible to
provide the audiences in different
domains
some additional works need to be
arranged and execute and there are some
practices what we understand uh what
need
to play their role and make

methodological work methodical work
and our work with some of our
disciplines
to make this knowledge accessible for
anyone with any level of expertise in
the field
yes it's a technical topic that we're
studying
and so there's a lot of work that goes
into
getting a working understanding and so
how are we going to scale that
and make that accessible stephen
yeah i think one other aspect is trust
you know
there's um they say in community
development you you can
well this one approach i really like is
that you move at the speed of trust
i think that's quite a nice thing and a
non-profit
shows a sort of commitment to values
which
you know implicitly is kind of there in
an academic organization
but the thing with academic
organizations there's certain
limitations
in that scope where a non-profit space
does offer a sort of a more pluralistic
flexible space as well
but again making sure that you can
establish that trust
in the values which i think the
commercial sector
um even with people like google who
said a lot about the values early on
then

it starts to you know so i think that is
an important
feature that you know is is part of this
process
cool thank you
how does the non-profit registration
update the phenotype of our lab
well again using active inference as a
framework for understanding this
we can break it down in terms of the
affordances that were provided
as well as the new niches which can be
physical
economic informational the new niches
that we're able to access
so we're better able to scaffold our
long-term partnerships
we're able to engage in new kinds of
interactions
like collaboration with other legally
recognized entities
or support of various kinds will be able
to
apply to new grant programs
as lead or as support will also be able
to
potentially support lab employees in
a sort of annualized way or in a
project-based way
and it will overall just help us serve
through deep time as we
provide improved onboarding and
accessibility efforts
so anyone who's interested in
this set of fields or any other field
that they think is going to be relevant
here
anything related to non-profit

management or
lab management research program
management making grant applications
some of the more classical areas of
interest for non-profits
that are research and technological as
well as
some of the more novel or online
characteristics that are important like
crypto economics
nfts uh crowdfunding token engineering
graphic design website development tool
development we'll talk more about some
of these things
later and a special note
for this second semester and then anyone
can provide a thought
is uh we're planning a lab advisory
board
we're planning to arrange this lab
advisory board
to start operations during this second
semester so
the rest of 2021 and we welcome
the suggestion or invitation to
individuals with high expertise in their
domain
people who would like to be part of our
advisory board
and some of the following areas of the
lab activities will be
supervised and catalyzed by the board
especially
legal education governance methodology
research strategy application slash
innovation
and really anything else so this again
helps us anchor out by connecting with

external

stakeholders and providing people who maybe they're not going to come to the dot edu meeting every week

but maybe once a quarter the uh the planets align

and we're able to interface with this person and so that can be hopefully a win-win interaction so if you're listening and

you think this is you or you know somebody who would be cool to invite onto the board just

let us know about that any thoughts on this or we'll continue on to the organizational units yep scott so two thoughts just on the exempt organization

uh status it's great congratulations on that

one is that the advisory board can also be a source of funding and other resources and network obviously and so constructing an advisory board and looking for those folks

i think that my impression is again from the conversation i'm having with finance folks etc is that there's going to be a lot of interest

uh people are looking at the framing and

i think getting quite excited about it

so there may be a lot of

action there and you can have different advisory boards multiple

boards as just as a structural matter

just with different functions etc

and you can also one of the things to

think about in the beginning is
sometimes having the advisory board
be a two-year stint or a three-year
stint or something so you can naturally
roll people off if people kind of linger
too long
or if you want to freshen it up so kind
of establishing that at the beginning
is something to think about keep it
fresh um
one other thing if the um if your
exempt organization purpose is includes
educational
then there i recall that there's
authority out there in private letter
rulings where educational institutions
were permitted to engage in standard
setting
which gets really interesting one of the
things i've told people involved in
education
i just spent time with some k-12 folks
last week
is that when there are standards out
there in the world education is about
teaching the standards so if i go to
learn french or chemistry i learn the
standards of french and chemistry in
school
where there are no standards then the
standard setting is done by curriculum
and so one of the things this teaching
that you're doing here is
also standard setting because you're
teaching framings
where in spaces where there may be none
and so it's something to think about
that the function of this lab may be

both educational
and standard setting and just to be
aware
that a lot of the outputs may feed into
standards efforts
as these there could be standards of
business operations
law technology and society that can come
out of this work
and so that's something to be aware of
and to
be intentional about if we start to go
in that route and making sure that
the deliverables that are intended to be
standards have the maturity
and the um exposure so that they can
have a chance as standards it's
something there's a real
opportunity in these new spaces these
new framings to pursue that
as well and under your auspices of your
501 exemption
thanks scott very nice stephen
yeah just further into the the
the finding standards and also
looking for processes where you've got
developmental
aspects and that's so this might be
where there's a kind of a couple of
streams going on because
um in a similar way to a lot of
non-profits that work in highly
innovative developmental spaces
you've got elements of that here
unusually for
um because it's not just the material
within a very standard framing the
vehicles

being developed um at the same time so
developmental evaluative approaches
principles focused evaluative approaches
could be

um which are actually very new in the
non-profit world which have taken
that have grown massively might really
be good for certain aspects but there
may be other aspects which start to get
locked down

in more ready already established
processes

and so that could be really really
useful

and we'll come back to this idea of
standards

in the next section which is the dot edu

so it's our first organizational
unit because learning is really

at the heart of the actin flab

in edu the goal is to create a
participatory and dynamic active
inference body

of knowledge and this is really drawing
on

alex and yvonne's experience with the
systems engineering

body of knowledge which isn't just a
reference book it's not just a wiki it's
not just the certifications not just the
standardization it's actually

this way of thinking about how these
pieces interact

and support everything from career
development to

people who are just curious to
conferences all these different

moving parts that play a role in that

living body of knowledge
so we'll provide an update here on the
main
efforts of edu which are
the ontology project with a working
group as well as a
uh an ontology in progress document and
then we'll talk a little bit about the
next
steps but any overall thoughts alex on
what body of knowledge means or how edu
helps us get there
i also want to highlight that a lot
educational activities in
priority and we understand that only
doing some projects with some new people
in the field
could lead us uh to have broad community
people who can work
in a project with active inference
that's why we are trying to approach
this
very fundamentally and go in through his
hard way
through methodological ontological works
and
always wet hard stuff
yep totally agreed so here's
uh yeah steven go ahead
yeah one part that i think is really
valuable with the
the body of knowledge is this idea of
the system of interest
which i know alex is you know they've
got a really
well-defined and evolving approach of
using systems of interest
and how that interfaces or with this

active inference
approach to a bounded
system or object and i think that has
proved to be sort of a consistency which
can bring things into the real world
in these participatory ways beyond say
purely
modeling in a more academic context so
i think that's been consistent and is
is a certain rigor that opens a lot of
good questions but i i'm really excited
by that
here's a figure that we've seen several
times it's a
spectrum that is describing
a uh sort of from the least formalized
forms of ontology on the left side to
the most formalized
forms the most expressive informal forms
of ontology on the right side
and knowing that ontology which is
how the world is studying the world and
how we frame it
for active inference we knew that was
gonna be a challenge but really the hill
that we had to climb
and we started in the first quarter of
2021
right when our lab begun with compiling
just a terms list
just what terms were relevant what were
we seeing come up
what were terms that people were
thinking were relevant for understanding
papers
and we're going to be working our way
towards the right side of this continuum
by introducing increasing levels of

annotation and formalization
right now what we're working on is
updating
our terms lists which we'll show in the
next slide
as well as introducing references and
definitions
so first just where have other people
defined a term in text and then
eventually we'll move towards creating
our own synthetic definitions
and then also introducing translations
and translations into natural human
languages as well as
translations eventually into
sulo sumo suo kif
which is if people want to check that
out that's really what the ontology
working group
is doing they're this train scooting to
the right
day by day uh trying to understand
this best-in-class way of working with
ontologies that adam peace and others
have developed and so we're working
through the textbook the ontology
textbook which everybody's welcome to
learn by doing as we work through this
textbook
and figure out what sumo is and we'll
use that
to start to develop computational ways
of thinking about the terms
that are relevant and along the way
we're going to be
structuring and providing all kinds of
interesting
ways of augmenting those lists and just

to
show what it looks like and then anyone
can
provide a comment and especially dave
who's done a ton of awesome work on
these
uh tabs in our working ontology document
which we'll
share with anyone who wants to see it
there's a few tabs there's
the core terms which are split into a
definitions and references
tab where we get up to five references
so published papers
that say you know term x is this
and just quote them what was it
described as in the text which
isn't always the clearest or isn't
always the most consistent across
definitions but that's the point we need
to know that
and then the second tab we have
translations
and at this point we have russian
portuguese
spanish french and italian are mostly or
completely
done but if you speak any other
language it'd be awesome to have your
translation effort
then we have a terms supplement where we
have kind of a
bunch of other terms that we think might
be relevant or might get upgraded to a
core term later or just are good to keep
on hand in
this broad dictionary that we're
building out

as well as in equations tab which dave
did some nice
copying over of equations because even
the equations the formalisms have have
somewhat evolved
or a letter will mean something in one
field but then it means something
different in another field or in a later
paper so we need these kinds of formal
ontologies because otherwise somebody
will say oh but g
meant this in this other paper and it
may have meant that
so how do we go from an internally
coherent like within the paper
coherent formalism to usages and
equations that are consistent across
papers
so that's what this work of connecting
the dots reading between the citations
that's what the ontology working group
is really working on
okay and the so again if you're
interested in this at all come contact
us and
join our ontology working group
the other dimension or project within
the dot edu
is this instructional and interaction
design
and the big question here is in the
medium term
how are we going to provide the
affordance for people to learn
active inference to update their
generative model through
action and inference about what active
inference is and how to apply it

and we're especially excited about
this dialectic between instructionism
and interactionism and how that might
relate to course design
and we talked about this in active
livestream 23
and dean and others had a lot of really
interesting things to say about this
but suffice to say that instructionism
is the idea that you're going to just
pass the telegram and it's going to be
received as an instruction
carried out like a cookbook
interactionism
is sort of where active inference takes
us which is
yes there is information passing between
the entities
however it's not simply that the
telegram message is being received
and the recipe is being carried out
there's something ongoing that's dynamic
that's being re-normalized at every
instance to for example
the surprise that the sensory unit is
experiencing
or the signals that the proprioceptive
aspects of the body are sending back
so what does that look like instead of
just making a sort of
list of terms or things you're going to
walk through and then after you've
gotten all these instructions you'll
know active inference
how could it be something that's
co-active with multi-agent sense making
and people working together and
interacting

to actually improve their model and then
just the last question and then stephen
is how do we provide applied active
inference education
for individuals coming from just so many
different
backgrounds that's going to be a
question we're always focused on
stephen
it might be as well when you look at
that interactionism
it might well be that when people use
or learn about how to interact with
active inference
how much of the active inference or what
kind of model
um would they maybe know about
so that that sort of raises that
question of
the different people's need to know
so to speak of what active influence
represents or is modelling could vary as
we enter into
some fields of systems engineering some
fields of psychology some fields of
uh so that could be quite interesting is
what kind of complexity or
scope of the model is needed to onboard
for a particular sort of
use case or discipline thanks
scott you know it got me thinking
um the last two you're coming and
stephen's comment
you know um the active inference the
framing
it allows a new matured perspective
on how information works and
one of the things we were on i was going

to call earlier today with a group that
just put together a
group field guide to misinformation
narrative campaign field guide it's
called
and one of the things we were talking
about was
that there's a sophistication we're kind
of wiring up the
collective human mind now and
active inference is kind of a necessary
prerequisite to being a good
neuron or a good participant
neurons the wrong word certainly um a
good participant in that system
and there's kind of a civic duty to kind
of understanding states
and understanding your state and other
states and i won't use the word
empathy or compassion but there's
definitely an intentionality and an
awareness that active inference lends to
systems
understanding of how communication works
and because
folks who i i s asserting these days
that the mind
doesn't exist in the brain it exists in
language and in culture
and the brain is just tuned into the
local mind
so we tune you know when you're babies
you that's why colonial
powers want to have indigenous children
in their schools they want to tune their
minds to imperialism
so one of the things that's kind of
interesting is

we're constantly mind tuning each other
and the awareness of other people's
internal states and the fact of these
processes
lends a new power and also new
responsibility
to shared communications and as the
species is getting
tighter packed and becoming in the world
is becoming smaller
informationally there's almost a
responsibility and duties associated
with
an active inference framing and it's
kind of a respectful
element to it or an awareness an
interdependence
element to it recognizing um this space
and what is going on in space and and
just to that point
when i first came into the space the
first thing i noticed was i said oh my
god my whole life
all i've been doing is contracts which
are synthetic markov blankets
right and it just there's awareness that
i had of what i was doing in the world
and how i could frame it
in the context of active inference that
really does
um lead to affordances social
affordances where there's a co-awareness
of the processes taking place
that can absolutely offer superior
pathways to governance because
information governance can be risk
governance
if properly approached and so this kind

of framing
can lead and carry the burdens of that
kind of thing because it's accessible
enough once the
basics are understood and if it's a
shared framing it becomes an
interoperable way of being
and that's part of what's missing in
information now is
interoperability um on cross-cultural
basis
thanks thanks scott so kind of
to close the section and then stephen
any last comments on edu
everyone is welcome to get involved with
this organizational unit through the
project-based learning
and the big umbrella that everything is
under is knowledge engineering
so whether you're translating languages
human languages computer languages
whether you're in the archive in stacks
looking for definitions and references
if you're working on curriculum
development and what that looks like
taking into account that whole
interesting dialectic
of instructionism and interactionism or
whether you're
in the ontology working group learning
ontology which is just such
a vast area but you're going to be
learning with other people who are also
learning whichever of these affordances
you take
hopefully it'll be something that you
can uh get a lot out of
by working with a group stephen

yeah building just one last point and
tied to what scott said is
states the idea of states um
and i think that's actually proved to be
where some of the last of the
uh terms list was being sort of worked
on as things that
related to states is a very much a jump
from what is being done
in most fields so you have even in
systems it's often
what's been moved around the dynamics
the
the type of objects but the state of the
object
is not normally exactly featured in the
same way and it's the same even in
psychology
someone's state might change but often
it says well they've been triggered it's
almost easier to say there's a trigger
and there's a thing that's triggered
someone
and talk about it in terms of objects
and stuff then actually
the state has changed so um
i think this is this is quite a hard
concept to get a head round but
comes up time and time again and of
course
inactive we partition states in a model
dependent way into internal and external
and then blanket states or boundary
states which first and then subdivided
into action
and sense but there we go we'll talk
more about that in the coming weeks
the next organizational unit is dot coms

which is communication
and dot coms has an internal and an
external goal
and function the internal goal is to
organize
all of the labs internal projects and
activities
that's kind of an organizer role and
then the external goal
is to carry out all forms of
communication with external entities
so if you email
activeinference@gmail.com
or if you contact us on twitter whatever
it happens to be
that interfacing externally is a
dot-coms
topic we're going to talk in these next
few slides
about some of the follow-ups from last
semester
and then we're going to look forward at
a few of the papers we've selected to
read in the coming weeks as well as give
opportunities
for places where people can make
suggestions for who'd be an awesome
guest
or what papers would be great to read so
the first
follow-up and uh dave if you want to
add any comments on this you'd be more
than welcome on
june 21st near the end of last semester
we had
our first applied active inference
symposium with carl fristen
and we had three different sessions each

about an hour long
carl was just such an incredible guest
and really generous with his time and
his answers
super educational and you can check out
the bitly link
or you can go to our youtube page where
you'll find these videos
and in terms of a follow-up something
that dave suggested and took the
initiative on
is to generate a transcript of the
proceedings of this symposium
and then we'll curate that add in the
references
clean it up and then hopefully pending
everyone's approval
have that transcript as a citable
preprint so instead of oh there was this
one video that one time where i think
this person said this
but maybe i'm not remembering it or how
many times does somebody say that and
then
it's like a tenth of them will go to the
timestamp and then a tenth will really
listen to what it says well
we're going to have a text
representation that we can then
translate
we can connect back to the formal
ontology and we can just
contribute to this knowledge
accumulation and especially some of his
answers which are not
written out anywhere else it'd be really
unique and important to have the
information

in this way so that's the follow-up from
the fristen symposium
this short link here which i'll also
post into the live chat
is the short link where you can always
go to
understand what is going to be happening
in the upcoming weeks with the streams
in terms of the papers that we're
reading
so here we are um in the beginning of
august
and in the first two weeks of august
we're going to be reading this recent
paper bayesian mechanics for stationary
processes
that's a pretty technical mathematical
paper and then in the following weeks
we're gonna um have authors on
discussing everything from
industrial engineering to
non-equilibrium thermodynamics
in late october and we also haven't yet
selected the papers for 28 29 and 30.
so three papers that we're going to be
able to
choose so if anybody has a paper that
they would like to read we have a list
of unplaced readings
in this document so you can scan through
that list
but if you have another paper you'd like
to suggest we're also always open to
that
and especially if it's your paper or if
you can contact the author and get them
to join us
for those weeks then that's bonus points

and this first tab of the spreadsheet is
our
tuesday at 7 pacific or 1400
gmt it's a participatory group
discussion
so it's best if you've at least tried to
read the paper
in terms of scanning through it but no
need to fully understand it
no need to be an expert we're all just
going there and
helping each other do sense making on
these really interesting
papers so please feel free to get
involved if you're curious about
live streaming or learning science out
in the open
however it presents itself to you
hopefully this is an affordance for you
to get involved with
live streams and with actinlab so
that's our usual
tuesday live stream series so at this
same exact time that we're doing
this live stream except other than the
four
round table meetings in the year every
other week we do a journal discussion
dave yeah have we ever looked at a paper
by martin butz
uh in germany uh toward a sub-symbolic
theory of computation
i'm really enjoying that i don't think
so
let's send you the link yeah great cool
so as far as what are the affordances
when we want to discuss a cool paper
so some papers we bring on to this

tuesday live stream we're going to
unpack it for two weeks in a group
hopefully with the authors and we're
also going to do a zero video
as a lab where we unpack the paper and
kind of
give this uh summary and also background
and context on the paper
but we also have the affordances of
these other series of streams
and just to kind of give a flavor and
also get people excited for
what is upcoming let's talk about a few
guest streams and model streams that we
have coming up
so this is on august 9th just next
monday
we're going to have mark miller is the
primary presenter
along with julian kiverstein ines
hippolyto and lars
sanfed smith the guest stream is going
to be called
exploring the predictive dynamics of
happiness and well-being
and just to read the first sentence of
their abstract
in this talk i will present some of our
recent work on mental health
and well-being using the active
inference framework
i i want to read more but you're going
to have to go listen to this
guest stream because i know it's going
to be a great one so that's
an example of a guest stream is some
authors want to present on
ongoing or recent work or they want to

just raise
up some perspective and we can just
schedule any time
that works for them you know 48 hours
later or three months later
the guest stream is kind of our wild
card where any
thing that somebody wants to present
that's related to active inference we're
always happy to give the venue for
and another upcoming event this is going
to be on august
19th so just in two weeks is going to
be model stream number 4.1 with
bert devrie this von der lar and dimitri
bagaiv
and it's going to be implementing active
inference by message passing in a factor
graph
and this is related to the work of the
bias lab
which is producing this forney lab julia
toolbox for factor graft
based probabilistic programming so this
diagram on the upper right is related to
what they're going to be talking about
so if you see that you think that looks
so cool or
how is that related to active inference
this is a really
powerful tool that these fellows are
working on so we're all excited to learn
more about that and maybe implement it
in some
cases so that'll be a great model stream
that's just
some of the kinds of papers as well as
the

guest appearances that we get to feature here

if you want to give a guest stream on your research or

if you have a suggestion for somebody who would be great for a guest stream or math stream or model stream

we're always open to that and bonus points if you can reach out to the author or if you are the author so that we can know that it's going to actually

happen and right now we're planning any kind of event and presentation for the rest of 2021

you know if it is a date in 2022 that you need to set aside let's do that too but we want to make the rest of 2021 really

powerful and really accessible to a lot of different backgrounds

so if you're interested in doing a guest stream or suggesting just get in contact with us anytime

we'll also in this dot coms section give a shout out to cognitio 2021

the website is here this is going to be an online conference

in the middle of october so only about a couple of weeks away and there's great co-organizers they just updated the website to reflect

some of the speakers and also several of our lab participants are going to be presenting research at this conference so if you're curious about active inference and collective intelligence

and also a lot of social topics this is
going to be an exciting
conference so just to close out
in dot coms and then anyone can provide
any thoughts
there are many affordances for
participating in dot coms we
welcome people who have been involved in
the lab already or who want to get
involved
with leading and growing the actin flap
accounts on social media
people who want to contribute or improve
the state of our current
series which are the podcast which is
primarily blue's work as well as the
live streams anyone who's interested in
introducing new series and formats
in compiling and curating the research
products of the active lab participants
we've already had multiple conference
presentations
papers where actinflab was written as an
affiliation
for an author people who want to help
onboard new participants in live streams
so they understand
how to set up everything from their
audio and visual to conceptually what
should one do to prepare
designing events blog short writing
tutorials
anything else that strikes your fancy so
hopefully if you're excited about
communication and or active inference
one of these
opportunities is right for you any
last comments on comms before we go to

dot tools

all right our third organizational unit

of three is dot tools

the goals of dot tools are to uh

again kind of like how dot coms had an

internal and external facing side

dot tools has sort of a a current and a

future facing side

our current facing side is to enable

effective tool use

and instrument use for all active

inference lab

processes so whether it's the video chat

platform we're using or live streaming

those are tools that we're using so we

want to use them skillfully

but also a little bit more broadly dot

tools is interested

in exploring and designing affordances

for our niche

resulting in effective action as well as

innovations in tool development

and dot tools last semester had

many weekly meetings where resources

were shared

and a lot of needs and ideas related to

the systems engineering of tools

were discussed to kind of give the

overview this is a figure from our 2020

paper active inference and behavior

engineering for teams

where we framed the operations of a

remote team

in terms of the structure of active

inference models

which is these multi-scale nested markov

blankets

with sense coming into the agent and

action going out to an agent and
communication
is kind of this special case whether
it's human communication or
two apis communicating or a human in a
server
because one's action is the other one's
sensory state
and so that kind of focus
allows us to take a function driven
approach
for how we think about communication and
tool development
using active inference so i know that in
the
tools brainstorming group people were
coming up with just
many different avenues and getting a lay
of the land
in terms of like what does an active
inference tool look like does it
look like a news feed but the ranking
algorithm is different or does it look
like a physical object does it look like
a new way of interacting with an object
we already have
a lot of pieces there
any other thoughts on this before um
alex why don't you then give a overview
of the net hack challenge
uh scott you had a comment oh
hey i just wanted to mention briefly on
the tools one of the things that's
really interesting to me
and new to the active inference world
here is that
um the notion of how the tools reveal
differentials

and my work i was an attorney for three
decades
and worked in risk basically that's what
attorneys do we try to de-risk things
and it usually has to do with
differentials information differentials
understanding differentials
and i think one of the things is going
to be really interesting is as the tools
develop
how might they be used in risk analyses
in different
areas i think that there's the the
ability to
navigate and close differentials that's
implicit
in the active inference framing i think
is going to prove to be
exceptionally valuable in industries
that deal with risk finance and
other institutional construction so it's
just something i wanted to raise about
the tools that
right now they're used it seems like
their analysis of active inference
itself
but i think they're incidental benefits
of these tools they'll find
application in myriad domains outside of
academic interest in academia
where the academic interest in active
inference where there's interest in
de-risking
and the tools will be used directly to
facilitate the closing
of gaps of information that are
associated with risk thanks
yep very cool

one of the applied projects oh yeah
stephen go ahead
well i was just going to say in in terms
of the developmental nature as well i
think
you know the um in
instructionalism and um interactionalism
which has
been the developing theme i think you
could say in the lab over the last few
months
starts to now maybe revisit some of
those earlier papers you might say
okay so we've got action so we've got
the idea of information being
passed through or imprinted into
a blanket as kind of blanket states
which goes into become
internal states and you've got the
blanket somehow
you know impression in itself on the
environment so you've got this
this this question about um
okay so you traditionally we may have
just been looking more instructionally
as in information comes in information
goes out well now we're looking at this
more interactionist dynamic with the
system boundary
you've both got an information flow and
some sort of
action the actors doing it and that
that then also becomes interesting maybe
as we
look at tools this sort of question
comes up um
and even how to diagram that
is up for grabs a little bit at the

moment

but uh i just thought that really made
me think of that

cool scott yeah just one other point and

i think there's a lot of movement now

where from a graph theory perspective

the difference of nodes and edges

we spent a lot of human history

concentrating on nodes

and that's really associated with

physical space it feels like

the edges the interactions the

relationships and the ability for us to

describe

attributes of edges and relationships i

think is increasing

and it's increasingly part of how we do

identity

and other things that we rely upon in

moving through space

and de-risking and leveraging our

behaviors

so that stephen's coming and that notion

of tools

really um highlighting the space between

nodes

and really allowing us to start to parse

relationships and parse the nature of

interactions and the

nature of state changes and things like

that what's really going on dynamically

and the bottom line is i think it gives

us a lot more solution space

with within which to think together

um and so that parsing of interactions

allows us to alight in that solution in

that

middle space between nodes and to

explore together with other people
especially when there's awareness of
that
and so it's very similar to what's
happening now i think in the
word misinformation is beginning to
attenuate
and people are starting to realize that
we just have a lot of information out
there and there's all sorts of context
and
and content and intention and all sorts
of things
so we can't just say misinformation or
information
and similarly the relationships need
parsing
the subtlety and nature of the
interactions of different nodes
in systems i think is going to be
revealed increasingly
by awareness of the attributes of the
relationships that are going to be re
helped to be revealed through active
inference analysis thanks
cool all right um alex
on to the net hack challenge so
all well and good but then how do we
really apply
these techniques and how will we give a
example
that amazes people what do you think
about this alex and what's the
vision here
yeah thanks this
challenge is looks really interesting
just because of its nature about that
legendary game

and uh a lot of people with
different ages backgrounds know
something or maybe even play from one
side from another side
it's really one of the difficulties
most difficult
[Music]
games in terms of solved by computers
and
by machine learning techniques so
and also for now it's interesting that
challenge
is hosted on the
on research conference and so
there are some connections between
possibly smart agents in this game
and connections with neuroscience
and in terms of complexity
and the visibility and we just have to
try
to show up some real life examples
of applying active inference agents to
some tasks
because even in our streams we saw some
examples
of real life agents which operates in
complex environment
so we know we have proofs that it could
be
possible on that technical examples
so we are trying to invite everyone from
the community
who are interested in machine learning
techniques and
active inference approach trying to
solve this task i don't know how
really hard it will be but at least it
should be fun

and could be first real life
applications
for with programming code of
active inference models
scott i have a question
um when the when active inference was
first kind of being explored and
proposed
my understanding was and tell me if this
is accurate that it was being abs
proposed as a model for modeling
biological and cognitive systems things
that were extent
in the environment and see if that's the
case
one of the things i'm wondering about is
what are the differences
when we intentionally invoke and
apply the model in a system versus when
we observe it in an existing system are
there differences can we
mimic is it biomimicry for us to pick up
this framing
and use it and what are the challenges
we and alex question to you in your
experience so far in kind of designing
this where we're
desiring to create and and this goes
back to my comment about
contracts being synthetic markov
blankets we are creating
i never thought of them as that but it
was if i were thinking of him is that it
might have made me do better contracts
so what are some of the differences if
there are any
between observing an active inference
framing and creating an active inference

framing in a system
you want to give a thought alex
um yeah i have
a lot of questions around this too but
uh watching latest model streams
on our channel uh brings
more perspectives and understanding what
need to be touched
uh to do it
it's it's almost like when we're talking
about a system that we're making
measurements
from we're doing inference on the
dynamics of
a system in the world and then when we
design
it does rely on eventually measurements
and inference to make sure everything is
going down the way that we want it to
but actually
it's a creative action that we're
undertaking to truly
design something and so it's a
interesting question especially in the
purely software realm
do we just port over the same exact um
sort of spatial biological or sensory
affordances
or are there new ways that
a biological system wouldn't have access
to like long-range communication or
some type of internal model tweak that
just doesn't really exist in biological
systems
so great question and uh we'll have the
nethack challenge team
come back to give you some thoughts
later dave

oh you're muted dave on the other side
okay i still can't hear you but when you
figure it out we'll come back to you but
yeah
um that's the net hack so
it's happening from uh
right now until october 15th so if you
want to get involved in nethack
challenge this would be the right time
to do it
and the skills as laid out in our
discord and also
on that previous slide especially if
you're interested
in learning and applying machine
learning
hacking gaming all these skills are
important this is not like tetris or
pac-man
the game itself is actually uh something
that has to be played through
because there's it's a challenging game
and um it should tell people something
that go
chess starcraft these games have been
played to extremely high level by
just off-the-shelf neural network
algorithms
and nethack hasn't and so that's
sort of the barking dog you know the dog
that didn't bark
it should let people know that it's the
type of problem
maybe that integrates multi-level
planning or
symbolic or discrete elements those are
the types of problems that
yes we can recognize the dog in the

image but can we play nethack and the
answer is no
also there's a special prize category
for people who are using a non-neural
network
so all things to think about um also
other affordances in dot tools
uh people who are interested in applying
active inference through
product research and engineering and
development that's
an availability alex will also
be uh helping us through a systems
engineering
project based learning process and of
course anyone who wants to help us
develop and improve our
tool usage any thoughts on the systems
engineering
uh side alex
okay um um
okay now dave now dave yeah can you hear
me now yep
yep oh good okay uh yeah my actually my
my major question is whether anything is
going on
with rosetta hub have you heard from tim
at all it seemed like he vanished he's
in the game
he's in the game you got your commute
because you're on a serious feedback
on a serious feedback loop
say that again i had to mute dave just
because he was on a serious feedback
loop
um with rosetta and whatever else we use
for cloud
architecture for the ontology working

group will keep people informed on those channels

scott one of the things i think is kind of interesting here is the if and to the extent that um active inference is something that's naturally occurs and we can observe in systems whether they know they're doing it or not

which i i assert to people i say that you're doing active inference whether you know it or not and there's articles on how humans use a beijing knowledge construction so those beijing notions are kind of embedded in a lot of human and other activities

to that extent it's kind of interesting because we

i always thought about this as bringing the active inference framing to the stakeholders

if and to the extent they're already doing active inference

it's kind of becomes a coaching element also

of revealing the skills that are already there the things that are already being done by the system

and then um tweaking them to maximize the throughput of whatever is desired by that organization

so it's a i hadn't thought of before is the

active inference as a naturally occurring phenomenon as an end as a synthetic phenomenon and we use the word artificial

but it's something we synthesize and how
what we're doing may be discovering an
improvement of an existing human
affordance
and and and not just human but living
affordance
and that puts a really different tweak
on it it's it's like um
you know you are here in a map you know
telling someone they're already doing
this
and let us show you how to improve it
it's like a capacity mining in a way
and and something we might think about
in the tools because i
hadn't thought about it that way before
but again the people doing it and
observing it where it occurs
among insects and cells which don't vote
and you know don't send out memos and
then
advocating i'm advocating at least for
among humans
who do vote and trip over their own feet
emotionally and politically all the time
and so it's kind of interesting if
active inference is a
um can be demonstrated to be
an efficient and or resource useful
or some other positive qualities because
of
nature engaging in it and that's a
demonstration of it
it offers an interesting rhetoric
to its instantiation in synthetic
context as well
that hey you're already half you're
already doing it why not do it well

kind of idea something to think about

cool scott thanks

steven

just to let you know i've i have heard

from tim in the sense that he is going

to be

sharing something on a session uh soon

he's just had a few

uh there's a few delays but um we've got

the support

from uh karen and latifah

and uh so i think progress will be

shared shortly

cool great um all right let's

look towards the future in our last

little section

the next steps for the lab which are

participatory so it's not going to

happen without people who want to step

up and step in and make it happen people

who want to act

the next steps are described within each

unit

so for edu comms and tools we talked

about

what kinds of projects we're going to be

doing this semester

and hopefully some of the affordances if

you want to help contribute or get

involved there

we're always looking to onboard new

participants

so if you're just hearing about this and

it's always the right time to get

involved

so contact us and then as we head into

our second semester of activity we

really want to focus

more on defining our governance
structures and on
exploring these new affordances that are
available to the non-profit
when we were working with the terms list
and then there were these interesting
discussions like
how will we decide what is a core term
and yes there's the conceptual question
of what constitutes a core term
but then there's the social question
about not everyone's going to have the
same
exact perspective on which terms should
or shouldn't be in the core terms
and maybe nine out of 10 people just
don't care and one person decides to say
yes or no on it
but that's not a governance structure so
how are we going to be able to work
truly as a collective in a participatory
lab
in terms of the products that we make
and the process
how we get there along the way so those
will be exciting
steps for our second semester
and we'll kind of just close out with
some of the questions that we always ask
about papers but our system of interest
is the lab
as organizers and as participants so we
want to know the same
exact things that we would ask about a
paper like
what do we want to learn how do we want
to apply it what would we
gain or what would be different if we

had a good understanding
what are the unique predictions and
implications how are we going to check
up along the way
what are the next steps what is our
policy selection
how are we going to reduce uncertainty
about the ways that we achieve our
preferences
what are the goals of the research and
then that remainder
the modulo which is our curiosity and
how are we going to be driven
by that differential to keep on
minimizing our free energy so that we
can continue
developing the lab scott
one of the things for us to be aware of
is we're
the lab is a machine to build machines
in a way right we're teaching the the
lab is
by doing the thing you're teaching or
learning or exploring
one thing to be really careful of is
when people come in that they don't get
confused by that
layering this is it happens a lot i'm
building a lot right now
of governance systems to build
governance systems
and so it's just something to be aware
of that when we're describing the
functions of the lab
that we're really explicit about the it
being
an instance of the thing it's developing
as well because that can get confusing

even though it's kind of there's a
naturalness to it you're living it's
like
gandhi you know be the future you want
to see kind of thing but or whatever the
quote was
but it's something to be aware of in
terms of the description
of the undertaking that we have this
yes we're doing active inference and yes
we're living active inference
so when sometimes when we're talking
about active inference stuff we're
talking about the organization
itself and sometimes we're talking about
active infant stuff we're talking about
the generalized concept of active
inference and
for us to they're not they don't need to
be separated
heavily but they do need to be juggled
because there is a layering that's a
that's
natural to us because we're working in
it now but a new entrant will get
confused by that
because they'll be confused on what lev
at what level they're engaging at
certain times so we need to be very
explicit
about the which level of recursivity
we're existing in when we're dealing
with something
in those multiple levels cool
very true so people can
give any last thoughts just raise their
hand but
otherwise we'll just say thanks so much

for participating this was a
great start to our second semester of
activities in the lab
any feedback or suggestions or questions
that people have we're always open and
the links here
contain how people can stay
informationally updated
stephen and then anyone else
that might be a nice linking to the next
uh live stream because i think that will
talk about
um what it means to segregate something
as an object
and when it means to maybe do more
you know predictive processing or and
then when you segregate things into
active
and sensory states so i think i think
that is a
an interesting question that as we get
more immersed
will be quite interesting is at what
point does someone
mostly need to know that there's
something happening at a very
embodied action sensory
process or where is it like the starting
point is
just that there's predictive processing
going on of some sort and that it's not
all
purely information coming in and then we
have the whole question about system of
interest again which
is always fun so thanks oh thank you
steven
any other last comments

uh yeah scott i just wanted to thank all
of you
for being so generous with your time and
your knowledge
and your ability and willingness to
engage to spread this this to me is a
very valuable
and um type of set of insights and it
allows for
a great deal of new innovation in a lot
of different areas
it goes to fundamentally to how we think
and so
i want to thank you for being beautiful
people
and beautiful in terms of your
generosity and your wisdom
so it really is a pleasure to be able to
associate with you and learn from all of
you so that's something i
look forward to in the next semester
thank you thank you scott
been great times so onto the paper
let's get on that bayesian mechanics
paper and start reading it and
highlighting every word we don't
understand yet aka all the words all the
equations but
um thanks everybody for participating
live and
for watching we'll see you in the coming
weeks with the guest stream with mark
miller at all on monday
and then following we'll just be back in
the rhythm of the papers
alright see you later